

# YANG YANG (3<sup>RD</sup> YEAR PHD CANDIDATE)

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**Anticipated Graduation Date:** 21 May 2028 **Citizenship:** China **Birth:** Sept 2001 **Version:** May 21, 2026

**Keywords:** Computer Architecture/GPU/Confidential Computing/Cryptography/Performance/Memory Technology

## STATEMENT

I am interested in **PERFORMANCE** issues in **GPU × TRUSTED COMPUTING × CRYPTOGRAPHY × MEMORY**

Recently, I have focused on performance issues in both conventional GPU environments (e.g., multi-GPU system [ISCA'25]) and secure GPU environments such as confidential computing (e.g., TDX-based H100 system [ISPASS'25, ISCA'26]).

For the next step, I will work on: (i) *scaling GPU-based CC*; (ii) *investigating the memory wall problem in GPU-based CC*; (iii) *making cryptographic protocols GPU-friendly*;

## EDUCATION

 **UNIVERSITY OF VIRGINIA, Charlottesville, USA**

**AUG. 2023 – NOW**

 **PH.D. in Computer Science**

**GPA:** 3.91/4.0

**Topics:** GPU × {Trusted-Computing, Cryptography, Memory}

**Advisor:** Dr. Adwait Jog

 **JILIN UNIVERSITY, Changchun, China**

**SEPT. 2019 – JUL. 2023**

 **B.S. in Computer Science**

**GPA:** 3.69/4.0

**Thesis:** The Design and Implementation of Binary Code Analysis Framework for NVIDIA GPU.

**Advisor:** Dr. Jingweijia Tan

## PUBLICATIONS

### [1] (ISCA'26)

**LÆGIS: Pinpointing and Addressing Performance Overheads of GPU-based Confidential Computing**

YANG YANG, Adwait Jog

(To Appear) In the Proceedings of IEEE/ACM International Symposium on Computer Architecture, Raleigh, NC, June 2026

### [2] (ISCA'25)

**NETCRAFTER: Tailoring Network Traffic for Non-Uniform Bandwidth Multi-GPU Systems**

Amel Fatima, YANG YANG, Yifan Sun, Rachata Ausavarungrun, Adwait Jog

In the Proceedings of ACM International Symposium on Computer Architecture, Tokyo, Japan, June 2025

### [3] (ISPASS'25)

**Dissecting Performance Overheads of Confidential Computing on GPU-based Systems**

YANG YANG, Mohammad Sonji, Adwait Jog

In the Proceedings of IEEE International Symposium on Performance Analysis of Systems and Software, Ghent, Belgium, May 2025

### [4] (SRC@CGO'23) [poster, before-PhD]

**Facilitating Profile Guided Compiler Optimization with Machine Learning**

YANG YANG, Xueying Wang, Guangli Li

SRC of IEEE/ACM International Symposium on Code Generation and Optimization, Montreal, Canada, February 2023

## EXPERIENCE

**NVIDIA Hardware Security Intern**

**MAY. 2026 – NOW**

**Santa Clara, California, USA**

**Manager:** Vidhya Krishnan, Ambarish Vyas **Mentor:** Santosh Ghosh, Moein Ghaniyoun

**Focus:** GPU Security, Confidential Computing and Cryptography (Homomorphic Encryption)

**Insight Computer Architecture Lab**

**AUG. 2023 – NOW**

**University of Virginia, Charlottesville, Virginia, USA**

**Advisor:** Dr. Adwait Jog

**Focus:** Architectural Support for Efficient and Secure GPU Computing.

- GPU-centric memory and storage system (DDR/HBM/SSD with UVM via PCIe/CXL/RDMA). [ISCA'25, ISCA'26]
- TDX-based confidential computing on GPUs. [ISPASS'25, ISCA'26]
- Cryptography (FHE/PIR) with GPU acceleration (work closely with Prof. Wei-Kai Lin).
- Efficient encryption for GPU-based heterogeneous memory system. [ISCA'26]

**State Key Laboratory of Processor  
Institute of Computing Technology, Chinese Academy of Science, Beijing, China**  
**Advisor:** Dr. Guangli Li

**JUL. 2022 – SEPT. 2023**

**Topics:** Compiler & Programming Systems

**Focus:** Facilitating Profile-Guided Compiler Optimization with Graph Learning

- Proposed a branch predictor using XGBoost based on compile time static analysis. [CGO-SRC'23]
- Utilize graph representations to build predictive profile-guided optimization framework and integrate it into LLVM.

**Emerging Technology Enabled Computer Architecture Lab  
Jilin University, Changchun, Jilin, China**

**FEB. 2022 – JUL. 2023**

**Advisor:** Dr. Jingweijia Tan

**Topics:** GPU  $\times$  {PTX/SASS, Reliability, Energy Efficiency}

**Focus:** Reliability and Energy-Efficiency of GPUs

- Soft error and hard error (e.g., process variation, NBTI) on FinFET based chiplet GPUs.
- SASS level analysing and programming framework for NVIDIA Ampere GPUs. [THESIS]
- Deep learning techniques for GPU power modeling.
- Explored how to reverse-engineer SASS and program in SASS.
- Instruction level under-voltage reliability of GPUs

## TALKS

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**LÆGIS: Pinpointing and Addressing Performance Overheads of GPU-based Confidential Computing**

- UVA CS SIG. May 2026

**NETCRAFTER: Tailoring Network Traffic for Non-Uniform Bandwidth Multi-GPU Systems**

- ACM ISCA. June 2025
- UVA CS SIG. September 2025

**Dissecting Performance Overheads of Confidential Computing on GPU-based Systems**

- IEEE ISPASS. May 2025
- UVA CS SIG. May 2025
- IBM Research (Virtual). May 2025
- Confidential Computing Summit. June 2025

## TEACHING

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**25 FALL @ UVA**, TA for CS 4444: Introduction to Parallel Computing (undergraduate)

- Co-Designed CUDA programming assignments.
- Delivered a lecture on GPU-based confidential computing.

**24 FALL @ UVA**, TA for CS 6354: Computer Architecture (graduate)

- Designed a pipelined RISC-V simulator in Python.

## AWARD

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[Travel Grant] ISPASS'25 ISCA'25 ISCA'26

[Fellowship@UVA] 2023

[Scholarship@JLU] 2019 2020 2021 2022

## SERVICE

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[Submission Chair] IISWC'26

[GPU-RIG] Coordinating logistics for GPU Research Interest Group across UVA SEAS in 2026.

[Artifact Evaluation Committee] IISWC'25

[SIG] Co-organizer of Systems-Interest-Group (SIG) meetings at UVA Department of Computer Science in 2024 and 2025.

## SKILLS

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[Languages] C/C++, Python, Assembly, Go, Rust

[Frameworks] CUDA, Pytorch, XGBoost, LLVM, TDX/SGX, QEMU/KVM, NVIDIA GPU Kernel Module, OpenFHE

[Software] Linux,  $\LaTeX$ , GNU compilers, GPGPU-Sim/Accel-Sim, NVCC/NCU/NSYS/NVBit, DRAMSim, Varius-TC, Z3

[Cryptography] AES (GCM, SIV, etc.), Merkle Trees, Public-Key (DH, RSA, etc.), PIR (LWE-based, FHE-based), ORAM

## PHD COMMITTEE (QUAL EXAM)

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**Chair:** [Wei-Kai Lin](#)   **Members:** [Sandhya Dwarkadas](#)   [Felix Lin](#)   [Adwait Jog](#)   **Announcement:** [\[Event link\]](#)